

Measurement of Inter and Intra-cycle Variations in Local Pulse Wave Velocity from Forward Travelling Pulse Waves

Rahul Manoj

Department of Electrical Engineering
Indian Institute of Technology Madras
Chennai, India
rahul_manoj@smail.iitm.ac.in

Raj Kiran V

Healthcare Technology Innovation
Centre (HTIC), IIT Madras
Chennai, India
rajkiran.v@htic.iitm.ac.in

Nabeel P M

Healthcare Technology Innovation
Centre (HTIC), IIT Madras
Chennai, India
nabeel@htic.iitm.ac.in

Mohanasankar Sivaprakasam

Department of Electrical Engineering
Indian Institute of Technology Madras
Chennai, India
mohan@ee.iitm.ac.in

Jayaraj Joseph

Department of Electrical Engineering
Indian Institute of Technology Madras
Chennai, India
jayaraj@ee.iitm.ac.in

Abstract—The local PWV and its pressure-dependent changes within a cardiac cycle are potential biomarkers for cardiovascular risk stratification. However, the influence of pulse wave reflections hampers the accuracy of local PWV measurements. We propose a method to measure the pressure-dependent variations in local PWV after minimizing the effect of wave reflections. The technique measures the pulse transit time between two simultaneously acquired and processed forward-running pulse waveforms obtained after wave separation analysis (WSA). A cross-sectional *in-vivo* study was conducted on 57 human participants (22 female) to compare the inter and intra-cycle variations in PWV based on pulse transit-time obtained from two measured pulse waves and two forward-travelling pulse waves. The reference PWV for comparison was obtained using Bramwell-Hill equations. The required signals – proximal and distal diameter waveforms from the carotid artery, simultaneously with carotid tonometry were acquired using an in-house developed bi-modal arterial probe. The carotid blood flow for WSA was recorded using an ultrasound imaging system. The reliability of measuring PWV was improved after WSA (coefficient of variation improved from 25% to 10% for PWV at the systolic level). The average PWV obtained before WSA and after WSA at diastolic levels were comparable with the reference PWV ($p > 0.05$). The average PWV obtained before WSA and after WSA at systolic levels were significantly different ($p < 0.001$); however, the PWV after WSA at systole level was comparable with the reference PWV ($p > 0.05$). The group average carotid pulse pressure derived from the mean values of PWV within the cardiac cycle, obtained after WSA (25.90 ± 5.88 mmHg), was statistically ($p > 0.05$) comparable to the carotid pulse pressure obtained using carotid tonometry-calibrated with brachial mean and diastolic blood pressure (26.74 ± 4.72 mmHg). Therefore, this proposed method has the potential to improve the measurement reliability of local PWV, aiding the ongoing global efforts of cardiovascular risk management.

Keywords— local PWV, incremental PWV, wave separation analysis, pulse transit time, wave reflection, blood pressure

This research was partially supported by Science and Engineering Research Board (SERB), Department of Science and Technology (DST) and Indian Institute of Technology (IIT) Madras under Institute of Eminence (IoE) funding from the Ministry of Human Resource Development (MHRD), Government of India.

I. INTRODUCTION

The velocity with which an arterial pulse traverses through the vasculature depends on both transmural blood pressure (BP) and the material properties of the vessel wall. The pulse wave velocity (PWV) measured over a long segment of the artery, such as carotid-femoral PWV (or regional PWV), is a known biomarker for cardiovascular risk stratification [1]. Recent advancements in sensing technologies have enabled the measurement of PWV over a short segment of an artery, such as the local PWV from fiducial points on a pulse waveform within a cardiac cycle. Local PWV has gained acceptance as a surrogate of localized stiffening of blood vessels and as an independent predictor of all-cause cardiovascular mortality [2]. Conventionally, the local PWV is measured from the foot of the pulse cycle, yielding a value at the diastolic pressure level. The incremental elastic nature of the central arteries dictates an incremental variation in PWV within a cardiac cycle. The maximum change in PWV within a cardiac cycle (change in PWV from foot to peak) has clinical significance as a potential biomarker for cardiovascular diagnosis [3], [4] and has applications in developing medical devices for calibration-free cuffless blood pressure measurements [5], [6], [7].

Several methods exist for estimating and measuring local PWV, namely – loop-based, transit-time-based, theoretical, impedance-based, and numerical-based techniques. A detailed review and summary of different approaches may be referred elsewhere [2], [8]. For the loop-based methods, the assumption of a reflection-free linear region in the early upstroke of the pulse waves is misleading [9] and has been experimentally shown to have presence of wave reflections even as close to systolic foot of the pulse waves [10]. The inconsistencies in the values of PWV observed among the various methodologies is also attributed to the influence of pulse wave reflections [11], [12], [13]. Additionally, these methods, would yield a single value of PWV [14]. Therefore, these methods are not suitable for measuring intra-cycle variations in PWV. On the other hand, the transit-time techniques are the most direct form of PWV measurement. This method involves recording two simultaneous pulse waveforms from a proximal and a distal site with a known distance on a localized artery. The pulse transit time (PTT)

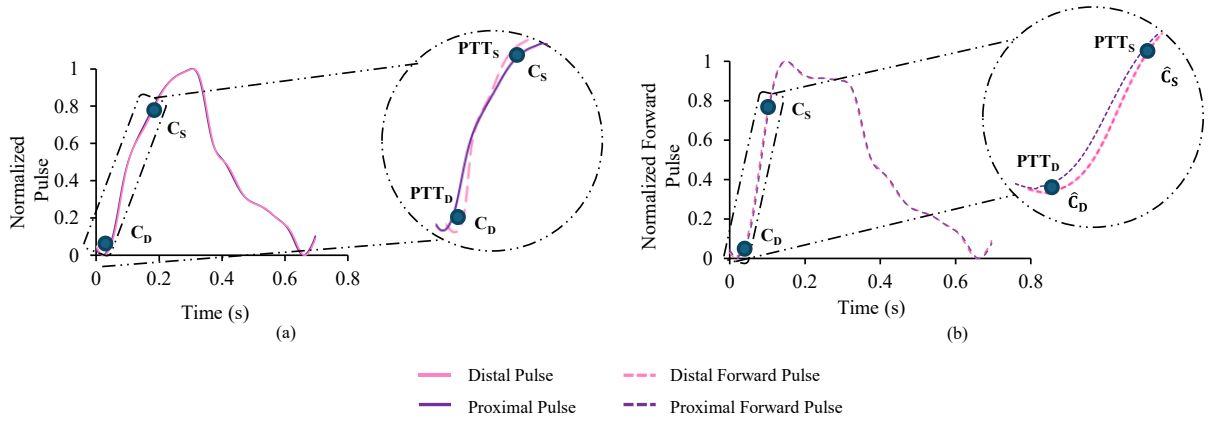


Fig.1 (a) PTT measured from the proximal and distal measured pulse waves, that illustrates an extreme case of crossing of pulse waves due to effects of pulse wave reflections, corrupting the measurement of PWV (b) WSA-derived forward pulse waves for the proximal and distal site, and the PTT measured from normalized forward pulse waves, devoid of any effects of pulse wave reflections.

between them is analyzed and combined with the distance measurement to yield PWV and its changes within a cardiac cycle. The measurement of PWV largely depends on the accurate description of the waveform morphology (Fig. 1) and the waveform features derived from it at the target measurement sites. The waveform features from both systolic peak [15], [16] and foot [10] are influenced by pulse wave reflections, modifying the pulse morphology, hindering the reliability of measuring local PWV and its pressure dependent variations.

Quantifying the pulse wave reflection and separation of forward-backward pressure waveforms has been an active research topic for several decades. A reliable method to quantify reflection is Wave Separation Analysis (WSA) [17]. This requires that the pressure waveform and flow rate waveform be measured, ideally from the same arterial site. In this work, we propose a method to measure the pressure-dependent variations in the local PWV by combining the advantages of the transit-time approach with WSA. The PWV is measured from the PTT between forward pressure waveforms obtained from the proximal and distal measurement sites after WSA. A similar approach for improved estimation of carotid-to-femoral PWV was demonstrated using arterial tube-load model-based wave separation algorithms [18], however, the applicability to local PWV or validation with theoretical PWV was not shown.

II. MATERIALS AND METHODS

A. Principle of Measurement

The local PWV calculated from the two-point method is based on the measurement of the PTT obtained from two pulse waves, one from a proximal point and the other from a distal point on the target artery as in (1),

$$C(t) = \frac{L}{PTT(t)} \quad (1)$$

where, t is varied from diastolic point to systolic point within the systolic phase of the pulse waveform. $C(t)$ is the PWV, with its incremental pressure-dependent variations within a cardiac cycle. L is the measurable anatomical blood vessel length between simultaneously recorded pulse waveforms [2]. The proximal and distal pulse waveforms for PTT measurement are the simultaneously measured pulse waves (before WSA) or forward-travelling pulse waves (after WSA).

PWV, before performing WSA, is labelled as $C(t)$ (refer to Fig. 1(a)) and after WSA as $\hat{C}(t)$ hereafter (refer to Fig. 1(b)).

The separation of forward-backward pressure waveforms using WSA is obtained by combining the water-hammer theory and transmission line electric circuit analogy to hydraulic circuits [19] as,

$$P_{F,prox}(t) = P_{prox}(t) + Z_C Q(t) \quad (2)$$

$$P_{F,dist}(t) = P_{dist}(t) + Z_C Q(t) \quad (3)$$

where, $P_{prox}(t)$ and $P_{dist}(t)$ are the simultaneously recorded pulse waveforms. $Q(t)$ is the flow waveform recorded from the same arterial site. $P_{F,prox}(t)$ and $P_{F,dist}(t)$ are the derived forward components of respective pulse waveforms. Z_C is the estimated characteristic impedance of the arterial vessel where measurement is performed. The WSA is either performed in the frequency domain or the time domain [17].

The carotid ΔP is calculated from the mean values of PWV (C_{mean}) obtained before and after WSA, combined with the BH equation as in (4),

$$\Delta P = 2\rho \times C_{mean}^2 \times \frac{\Delta D}{D_D} \quad (4)$$

Where, ρ is the density of blood, ΔD is the maximum wall distention, and D_D is the end-diastolic lumen diameter of the carotid artery.

B. Instruments Used for Data Collection

A modified version of the in-house developed bi-modal arterial probe [20] combining two custom-made single-element focused ultrasound transducers (center frequency: 5 MHz, diameter: 5 mm, spatial angle $< 1.5^\circ$, center-to-center distance between transducers: 35 mm) and tonometer sensor cartridge (SPT-301, Millar Instruments, USA, Sensitivity: $5\mu V/V/mmHg$, Range: -50 mmHg to 300 mmHg), was designed to acquire the radio frequency (RF) echoes simultaneously from two locations on the carotid artery vessel walls, along with carotid-tonometry pulse waves. The instrumentation was designed for multi-modal signal acquisition in a synchronized manner for beat-to-beat evaluation of both processed diameter waveforms and tonometry pulse waves. The details on the hardware and software architecture of the bi-modal probe are referred to elsewhere [20]. The signals obtained from the bi-modal arterial probe were used to derive the PWV using the two-

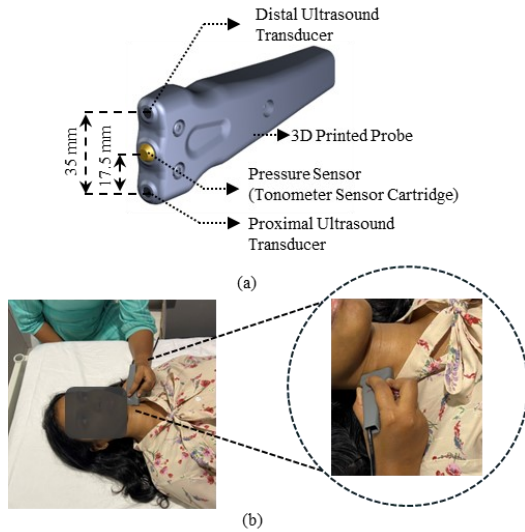


Fig.2 (a) Bi-modal arterial probe, and (b) the position at the common carotid artery.

point transit-time method and the reference PWV using the BH equation. Fig. 2 depicts the components of the bi-modal arterial probe and its probe positioning at the carotid artery of a participant. The brachial artery BP, used to calibrate the tonometry pulse waves, was measured using an oscillometric BP monitoring device (SunTech 247, USA).

The Doppler flow measurements were recorded using the duplex mode from an ultrasound imaging system (Sonix Touch+, BK Medical, USA). The imaging system consists of a linear array probe (center frequency: 10 MHz, pitch: 0.3 mm, depth: 4 cm). The duplex mode with a frame rate of 40 Hz pulsed wave Doppler was used to record continuous flow velocity waveforms as a video file in .avi format. The recorded video files were converted into time series flow velocity waveforms using an open-sourced software [21] in MATLAB® and saved as .csv files for further analysis. The required $Q(t)$ was obtained by combining the vessel cross-sectional information obtained from the bi-modal arterial probe with flow velocity obtained from the Doppler ultrasound.

C. Study Design and Protocol

The cross-sectional study comprised 57 participants (22 female) from the age group 20 to 56 years. The study was conducted at the Indian Institute of Technology Madras (IIT M). The study design and protocol were approved by the Institute Ethics Committee of IIT M (IEC/2021-01/JJ/07). The participants were informed of the study outcomes, and written consent was obtained for their voluntary participation. The data collection for the study adheres to the latest guidelines of the Declaration of Helsinki for medical research involving human subjects.

Before the study, the participants completed a questionnaire on their medical and genetic history and current medication. Anthropometric measurements were recorded, including height, weight, and Body Mass Index (BMI). The study was conducted in a temperature-controlled room ($\sim 24^\circ\text{C} - 26^\circ\text{C}$), and the participants had to lie in a supine posture for the entire study duration. The measurements start with brachial artery BP, recorded using an oscillometric device. The measurement was repeated thrice, and average systolic blood pressure (SBP), DBP and MAP values were used for

analysis. For the carotid artery flow velocity measurements, the linear array probe of the ultrasound imaging system was positioned and oriented on the neck region of the participants to obtain a clear image of the carotid artery in the longitudinal section. Appropriate gate width ($\sim 1/3^{\text{rd}}$ of the lumen diameter) and angle were configured to measure the Doppler flow velocity of the blood flow. The Duplex mode of the imaging system records the flow velocity waveforms for thirty seconds as a video file.

For measurements using the bi-modal arterial probe, the neck region of the participants was palpated to identify the location of the carotid artery. The probe was oriented to get sharp echoes from the near wall and far wall of the vessel from both sites. The probe was slightly appanated to capture the tonometry pulse waves with maximum amplitude from the carotid artery. Continuous RF data and tonometry pulse waveforms were recorded for thirty seconds for further processing. Multiple measurements of thirty seconds were obtained to select the best-quality waveforms for analysis. Both the measurements of Doppler flow and the bi-modal arterial probe were obtained sequentially.

D. Data Processing

Fig. 3 illustrates an overview of the algorithm and signal processing steps involved. The vessel walls were identified and tracked from the simultaneously recorded RF echoes for proximal and distal sites using validated tracking algorithms [22] to obtain continuous diameter waveforms. The proximal and distal diameter waveforms were nonlinearly scaled to pressure waveforms based on Hayashi-Kawasaki pressure-diameter relationships and calibrated with brachial artery MAP and DBP [23]. The tonometry pulse waves were linearly calibrated with brachial artery MAP and DBP to obtain carotid pressure waveforms. The Duplex mode video files with flow velocity waveforms (in cm/s) were processed using open-sourced software written in MATLAB® [21]. The obtained time series waveform was calibrated with the on-screen axis to equivalent units in pixels. The flow velocity waveforms were multiplied by the cross-sectional area waveforms to obtain the flow rate waveform (in ml/s) for WSA. All the processed pulse waveforms – proximal and distal diameter, tonometry pulse, and Doppler flow were up-sampled to 10 kHz and underwent the same zero-phase low pass Butterworth filtering (cutoff: 15 Hz, order: 3) to preserve the signal integrity across the waveforms. The sequentially obtained flow waveform is time-aligned with the simultaneously obtained proximal, distal diameter and tonometry waveforms by adjusting to the end-diastolic foot of all the waveforms. Five continuous cycles of diameter, tonometry pressure and flow are time-averaged for performing WSA.

The pulse waves were normalized from 0 to 1 for the magnitude thresholding-based PTT estimation. The time instant corresponding to 10% and 80% from the end-diastolic foot of both the proximal and distal pulse waveforms were calculated, and the temporal difference between them was measured as the PTT_D and PTT_S within a cardiac cycle. The corresponding C_D and C_S were calculated using (1). However, for reflection-free PWV, the forward-travelling pulse waves were derived from WSA for both proximal and distal sites, for PWV ($\hat{C}_D, \hat{C}_S$) calculation as illustrated in Fig 1. For WSA, the proximal and distal diameter-calibrated pressure waveforms and flow waveforms were transformed into their frequency domain components. The magnitude spectrum of

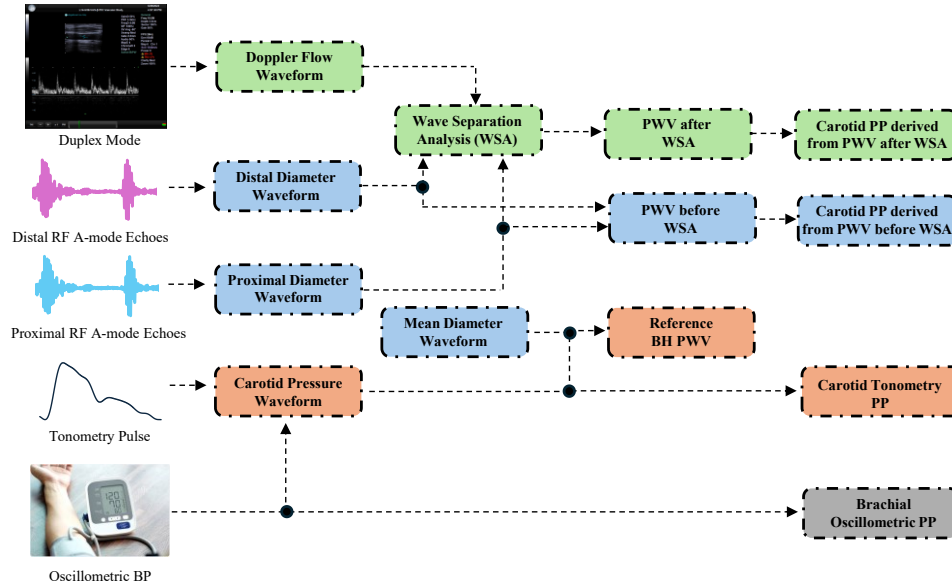


Fig.3 Data processing flow chart, illustrating the various algorithms involved.

the input impedance was calculated as $|P(j\omega)|/|Q(j\omega)|$. The Z_C was estimated as the average magnitude of the input impedance from the 4th to 10th harmonics [17]. The $P_{F,prox}(t)$ and $P_{F,dist}(t)$ were calculated using (2) and (3).

E. Statistical Analysis

All the group average measurements were reported as mean $\pm$ standard deviation. The variability in beat-to-beat measurements was analyzed using the coefficient of variation (CoV). The CoV was obtained as the ratio of respective standard deviation to mean, expressed as a percentage. The Box-and-whisker plots and t-test paired two samples for means were used to indicate the similarity or difference between estimated quantities. The level of significance of $\alpha = 0.05$ was used for all tests. A p-value < 0.05 confirmed a statistical significance to reject the associated null hypothesis.

III. RESULTS AND DISCUSSION

A. Participants Demography

A detailed summary of the participant's demography is presented in Table I. The study involved 57 participants; 38.5% of the total participants were female. The mean age of the participants was 32 ± 10 years. The participants had brachial artery SBP varying from 182 mmHg to 92 mmHg and DBP from 56 to 122 mmHg. No participants were under any prescribed medication during or immediately prior to the study period. Female participants were pre-menopausal and non-pregnant. Five participants were hypertensive; three participants had a habit of smoking. The data collection from each participant took an average of 20 minutes to complete.

B. Reliability of the Recorded Signals

The proximal and distal RF echo signals had a Signal-to-Noise Ratio (SNR) > 25 dB, sufficient for the vessel wall identification and tracking to obtain the proximal and distal diameter waveforms, with a resolution of $10 \mu\text{m}$ and a frame rate of 500 Hz. The diameter waveforms were continuous and quasi-periodic in nature. The average D_D among the study participants was 6.19 ± 0.96 mm, with a mean CoV $< 7\%$ and the average ΔD was 0.51 ± 0.16 mm, with a mean CoV $< 5\%$. The calibrated pressure signal from the tonometer sensor had

TABLE I PARTICIPANTS DEMOGRAPHY

Total participants	57
Male participants	35 (61.5 %)
Female participants	22 (38.5 %)
Age (Years)	20 – 56
Height (cm)	169.8 ± 11.5
Weight (Kg)	68.5 ± 8.1
Body Mass Index (Kg/m ²)	24.5 ± 5.1
Heart Rate (bpm)	70 ± 11
Brachial SBP (mmHg)	116 ± 16
Brachial DBP (mmHg)	73 ± 12
Brachial MAP (mmHg)	87 ± 13
Brachial PP (mmHg)	42 ± 6
Carotid Tonometry SBP (mmHg)	99 ± 13
Carotid Tonometry PP (mmHg)	26 ± 4
Peak Systolic Flow Velocity (cm/s)	60 ± 11
End Diastolic Flow Velocity (cm/s)	14 ± 4

an SNR of 20 dB. Continuous and quasi-periodic carotid pulse waveforms were reliably recorded. The pair of diameter waveforms (proximal and distal) and calibrated pressure waveforms were acquired synchronously using the bi-modal arterial probe for beat-to-beat evaluation of PWV.

The Doppler flow velocity waveforms obtained from the duplex-mode ultrasound video file have reliably captured the tri-phasic nature of the carotid artery flow velocity waveform. The waveforms were up-sampled to 500 Hz. The study participants' peak systolic and end-diastolic flow velocities were 60 ± 11 cm/s and 14 ± 4 cm/s, respectively. Upon combining with the cross-sectional area of the carotid artery, the peak flow rate was 21.6 ± 7 ml/s, and the end-diastolic flow rate was 4.5 ± 2 ml/s.

C. Wave Separation Analysis

The average pulse pressure of forward and backward pressure waveforms (ΔP_F and ΔP_B) among the study participants were 25.71 ± 6.04 mmHg and 13.05 ± 4.16 mmHg respectively. The group average reflection magnitude (RM), reflection index (RI) and reflection wave transit time (RWTT) were 0.55 ± 0.23 , $32.45 \pm 4.69\%$, and 68.50 ± 14.54 ms, respectively.

The WSA performed on carotid artery reports RM with a mean value of 0.68 ± 0.10 to 0.76 ± 0.10 and RI with a mean value of $40.0 \pm 4.0\%$ to $43.0 \pm 3.0\%$ from *in-vivo* study with

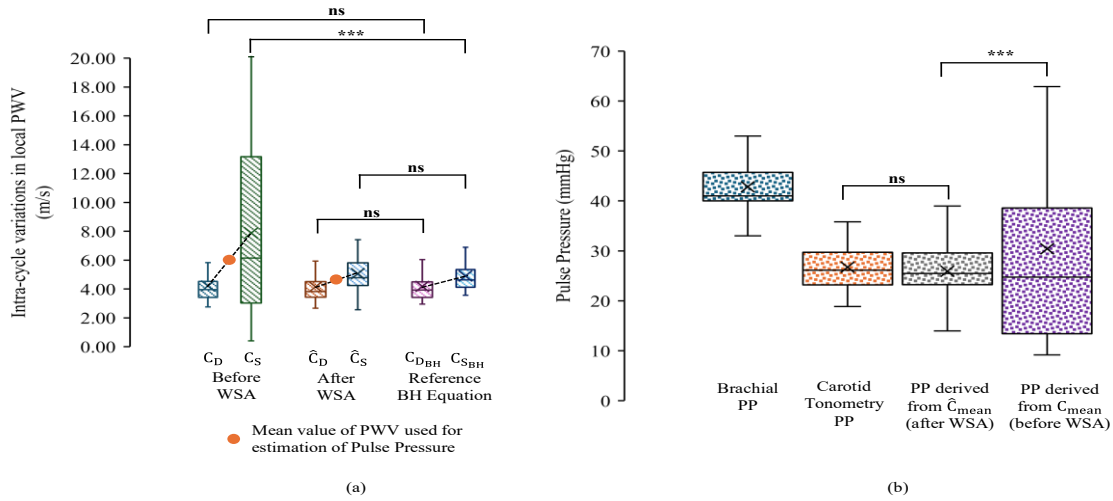


Fig. 4 (a) PWV before WSA and after WSA, compared with reference PWV obtained from BH equation, (b) derived estimates of carotid PP before WSA, after WSA and carotid tonometry.

an average age of 49 ± 17 years [24]. In the current study, with an average age of 32 ± 10 years, the mean values of RM and RI were lower than those reported with a higher mean age. The other studies on carotid WSA were our previous works using multi-Gaussian decomposition of pressure waveform [25] and multi-Rayleigh flow model-based WSA [26] performed on an *in-silico* database, which yielded values of RM, RI, RWTT in the ranges of 0.79 ± 0.08 , $44.23 \pm 2.58\%$ and 45.50 ± 5.56 ms respectively for mean age of 45 ± 10 years. These observations were concurrent with the group average values observed in this study after accounting for the lower mean age. The WSA is essential for obtaining the forward pulse waves used to measure transit-time-based PWV. Most studies on WSA are limited to aortic sites. Therefore, it was necessary first to quantify the reflections obtained from the carotid artery to show the performance of the WSA algorithms.

D. Intra-cycle & Inter-cycle Variations in PWV

Fig. 4 (a), depicts the changes in the average value of PWV obtained from transit-time-based measurements on pulse waves before WSA (C_D , C_S) and after WSA ($\hat{C}_D$, $\hat{C}_S$) within a cardiac cycle across the study population compared with reference PWV. The mean values of theoretical PWV from two points within a cardiac cycle (C_{DBH} and C_{SBH}) for the study population were 4.18 ± 1.06 m/s and 5.31 ± 1.13 m/s respectively. The average value of C_D was 4.20 ± 1.12 m/s and $\hat{C}_D$ was 4.15 ± 1.16 m/s, whereas the average value of C_S was 8.38 ± 6.45 m/s and that of $\hat{C}_S$ was 5.07 ± 1.45 m/s. There exists a statistically insignificant difference ($p > 0.05$) between the PWV at the diastolic point ($\hat{C}_D$ and C_D) and the C_{DBH} at diastole. However, there exists a significant difference ($p < 0.001$) between the mean values of C_S and C_{SBH} at systole. Whereas it was observed that the $\hat{C}_S$ is statistically comparable to C_{SBH} at systole levels ($p > 0.05$).

The average CoV for C_D was $\sim 8\%$, and C_S was $\sim 25\%$. From using forward-travelling waves for the measurement of PWV, the average CoV was improved for $\hat{C}_D \sim 7\%$, and that of $\hat{C}_S$ improved to 10% . The ranges of intra-cycle variations of PWV for diastole and systole levels were consistent with recent studies performed using ultrafast ultrasound systems [3], [4] and studies had commented on the effect of wave reflection could corrupt the measurements of PWV at systolic

levels. The pressure-dependent variations of PWV using the BH equations were explored, and it was reported that local PWV values were significantly higher for systolic measurements than diastolic measurements [27], in line with the observations reported in this article.

The pressure dependency of the PWV within a cardiac cycle is obtained by combining the Bramwell-Hill relationship with the Hayashi-Kawasaki empirical pressure-diameter model, as explored in [2], [5]. The exponential nature of this relationship is attributed to the incremental nature of the stress-strain curve of the arterial vessel wall. The elastin component of the vessel wall becomes the load-bearing component at a lower pressure level, whereas collagen fibers become the load-bearing component at a higher-pressure level. Similar trends of pressure-dependent PWV were previously reported in *ex-vivo* excised artery studies [28], [29]. With the recent advancement of noninvasive sensing technologies over the last few decades, the pressure dependency of PWV was explored for cuffless blood pressure measurement [5], [30]. In particular, local PWV and its pressure-dependent variations have been recognized as a measurable parameter for assessing both the functional and structural stiffness of target arteries [2]. More recently, the intra-cycle variations in PWV (ΔC) have been explored for their predictive and therapeutic applications [3], [4].

E. Derived Estimates of Carotid Pulse Pressure

Fig. 4(b) depicts the variations in the derived estimates of carotid PP from mean values of PWV before and after WSA. These estimates are compared with the carotid PP obtained from linear calibration of the tonometry pressure waveform. The carotid PP before WSA has a mean value of 30.41 ± 21.24 mmHg and is significantly different from the carotid PP obtained from tonometry ($p < 0.001$). The carotid PP, after WSA, had a mean value of 25.90 ± 5.88 mmHg and is comparable to the mean value of carotid PP obtained from tonometry ($p > 0.05$). The variability of the carotid PP after WSA is also in range with that of the carotid PP obtained from tonometry (Standard deviations: 5.88 vs. 4.72). The carotid PP obtained from the BH equation using the mean values of PWV within a cardiac cycle and diameter waveform as in (4) was a cross-validation of the proposed method to improve the measurement reliability for PWV.

IV. CONCLUSION

We highlight a reliable method for the measurement of local PWV that effectively minimizes the distortive effects of pulse wave reflections. Our proposed method, leveraging wave separation analysis (WSA) to assess the pressure-dependent variations in local PWV, demonstrates enhanced measurement reliability through the comparison of inter and intra-cycle variations in PWV among 57 participants. The study underscores the method's efficacy in aligning the measured PWV closer to the reference values, particularly after WSA, with a notable reduction in coefficient of variation (from average CoV of 25% to average CoV of 8% at systolic level) and statistically significant ($p > 0.05$) and comparable values of PWV at the systolic level with reference BH PWV (5.31 ± 1.13 m/s Vs. 5.07 ± 1.45 m/s). This method can potentially improve the measurement reliability of local PWV, aiding the cardiovascular risk management protocols and contributing to the global efforts towards more accurate and reliable cardiovascular health assessment.

REFERENCES

- [1] F. U. S. Mattace-Raso *et al.*, "Determinants of pulse wave velocity in healthy people and in the presence of cardiovascular risk factors: 'Establishing normal and reference values,'" *Eur. Heart J.*, vol. 31, no. 19, pp. 2338–2350, 2010, doi: 10.1093/eurheartj/ehq165.
- [2] P. M. Nabeel *et al.*, "Local Pulse Wave Velocity: Theory, Methods, Advancements, and Clinical Applications," *IEEE Rev. Biomed. Eng.*, vol. 13, no. c, pp. 74–112, 2020, doi: 10.1109/RBME.2019.2931587.
- [3] P. M. Nabeel, D. S. Chandran, P. Kaur, S. Thanikachalam, M. Sivaprakasam, and J. Joseph, "Association of incremental pulse wave velocity with cardiometabolic risk factors," *Sci. Rep.*, vol. 11, no. 1, pp. 1–10, 2021, doi: 10.1038/s41598-021-94723-2.
- [4] E. Hermeling *et al.*, "The change in arterial stiffness over the cardiac cycle rather than diastolic stiffness is independently associated with left ventricular mass index in healthy middle-aged individuals," *J. Hypertens.*, vol. 30, no. 2, pp. 396–402, 2012, doi: 10.1097/HJH.0b013e32834e4b75.
- [5] P. M. Nabeel, J. Joseph, S. Karthik, M. Sivaprakasam, and M. Chenniappan, "Bi-Modal arterial compliance probe for calibration-free cuffless blood pressure estimation," *IEEE Trans. Biomed. Eng.*, vol. 65, no. 11, pp. 2392–2404, 2018, doi: 10.1109/TBME.2018.2866332.
- [6] K. V. Raj, J. Joseph, N. P. M., and M. Sivaprakasam, "Automated measurement of compression-decompression in arterial diameter and wall thickness by image-free ultrasound," *Comput. Methods Programs Biomed.*, vol. 194, p. 105557, 2020, doi: 10.1016/j.cmpb.2020.105557.
- [7] P. M. Nabeel, K. V. Raj, and J. Joseph, "Image-free ultrasound for local and regional vascular stiffness assessment: The ARTSENS Plus," *J. Hypertens.*, vol. 40, no. 8, pp. 1537–1544, 2022, doi: 10.1097/HJH.0000000000003181.
- [8] V. Raj Kiran, P. M. Nabeel, J. Joseph, and M. Sivaprakasam, "Methodological and Measurement Concerns of Local Pulse Wave Velocity Assessment," *Med. Meas. Appl. MeMeA 2019 - Symp. Proc.*, pp. 1–6, 2019, doi: 10.1109/MeMeA.2019.8802181.
- [9] P. Segers, E. R. Rietzschel, and J. A. Chirinos, "How to Measure Arterial Stiffness in Humans," *Arterioscler. Thromb. Vasc. Biol.*, vol. 40, no. March, pp. 1034–1043, 2020, doi: 10.1161/ATVBAHA.119.313132.
- [10] E. Hermeling, K. D. Reesink, R. S. Reneman, and A. P. G. Hoeks, "Confluence of incident and reflected waves interferes with systolic foot detection of the carotid artery distension waveform," *J. Hypertens.*, vol. 26, no. 12, pp. 2374–2380, 2008, doi: 10.1097/HJH.0b013e328311cdd5.
- [11] P. Segers, A. Swillens, L. Taerlman, and J. Vierendeels, "Wave reflection leads to over- and underestimation of local wave speed by the PU- and QA-loop methods: Theoretical basis and solution to the problem," *Physiol. Meas.*, vol. 35, no. 5, pp. 847–861, 2014, doi: 10.1088/0967-3334/35/5/847.
- [12] R. Manoj, V. Raj Kiran, S. Ponkalaivani, P. M. Nabeel, M. Sivaprakasam, and J. Joseph, "Measurement of Local Pulse Wave Velocity: Agreement Among Various Methodologies," *2023 IEEE Int. Symp. Med. Meas. Appl. MeMeA 2023 - Conf. Proc.*, pp. 1–6, 2023, doi: 10.1109/MeMeA57477.2023.10171907.
- [13] V. Raj Kiran, P. M. Nabeel, J. Joseph, M. I. Shah, and M. Sivaprakasam, "Evaluation of Local Pulse Wave Velocity using an Image Free Ultrasound Technique," *MeMeA 2018 - 2018 IEEE Int. Symp. Med. Meas. Appl. Proc.*, vol. 3528725544, pp. 1–6, 2018, doi: 10.1109/MeMeA.2018.8438649.
- [14] B. Spronck *et al.*, "Heart rate and blood pressure dependence of aortic distensibility in rats: Comparison of measured and calculated pulse wave velocity," *J. Hypertens.*, vol. 39, no. 1, pp. 117–126, Jan. 2021, doi: 10.1097/HJH.0000000000002608.
- [15] P. Segers *et al.*, "Limitations and pitfalls of non-invasive measurement of arterial pressure wave reflections and pulse wave velocity," *Artery Res.*, vol. 3, no. 2, pp. 79–88, 2009, doi: 10.1016/j.artres.2009.02.006.
- [16] L. Marais *et al.*, "Arterial Stiffness Assessment by Shear Wave Elastography and Ultrafast Pulse Wave Imaging: Comparison with Reference Techniques in Normotensives and Hypertensives," *Ultrasound Med. Biol.*, vol. 45, no. 3, pp. 758–772, Mar. 2019, doi: 10.1016/j.ultrasmedbio.2018.10.032.
- [17] J. P. Mynard and A. Kondiboyina, "Wave reflection in the arterial tree," in *Textbook of Arterial Stiffness and Pulsatile Hemodynamics in Health and Disease*, vol. 1, J. A. Chirinos, Ed., Academic Press, 2022, pp. 169–194. doi: 10.1016/B978-0-323-91391-1.00011-X.
- [18] M. Gao, G. Zhang, N. B. Olivier, and R. Mukkamala, "Improved pulse wave velocity estimation using an arterial tube-load model," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 3, pp. 848–858, 2014, doi: 10.1109/TBME.2013.2291385.
- [19] N. Westerhof, P. Sipkema, G. C. V. Den Bos, and G. Elzinga, "Forward and backward waves in the arterial system," *Cardiovasc. Res.*, vol. 6, no. 6, pp. 648–656, 1972, doi: 10.1093/cvr/6.6.648.
- [20] R. Manoj, R. Kiran, P. M. Nabeel, J. Joseph, and M. Sivaprakasam, "A Bi-modal Probe Integrated with A-mode Ultrasound and Force Sensor for Single-site Assessment of Arterial Pressure-Diameter Loop," in *IEEE Medical Measurements and Applications, MeMeA 2020 - Conference Proceedings*, Bari, Italy: IEEE, 2020, pp. 1–6. doi: 10.1109/MeMeA49120.2020.9137189.
- [21] C. L. Coolbaugh, E. C. Bush, C. F. Caskey, B. M. Damon, and T. F. Towse, "FloWave.US: Validated, open-source, and flexible software for ultrasound blood flow analysis," *J. Appl. Physiol.*, vol. 121, no. 4, pp. 849–857, 2016, doi: 10.1152/japplphysiol.00819.2015.
- [22] J. Joseph *et al.*, "ARTSENS® Pen - Portable easy-to-use device for carotid stiffness measurement: Technology validation and clinical-utility assessment," *Biomed. Phys. Eng. Express*, vol. 7, no. 2, 2020, doi: 10.1088/2057-1976/ab74ff.
- [23] J. M. Meinders and A. P. G. Hoeks, "Simultaneous assessment of diameter and pressure waveforms in the carotid artery," *Ultrasound Med. Biol.*, vol. 30, no. 2, pp. 147–154, 2004, doi: 10.1016/j.ultrasmedbio.2003.10.014.
- [24] N. Di Lascio, V. Gemignani, E. Bianchini, R. M. Bruno, L. Ghiadoni, and F. Fata, "Effects of carotid pressure waveforms on the results of wave separation, wave intensity and reservoir pressure analysis," *Physiol. Meas.*, vol. 39, no. 11, 2018, doi: 10.1088/1361-6579/aae6eb.
- [25] R. Manoj, K. V. Raj, P. M. Nabeel, M. Sivaprakasam, and J. Joseph, "Arterial pressure pulse wave separation analysis using a multi-Gaussian decomposition model," *Physiol. Meas.*, vol. 43, no. 5, p. 055005, May 2022, doi: 10.1088/1361-6579/ac6e56.
- [26] R. Manoj, P. M. Nabeel, V. Raj Kiran, M. Sivaprakasam, and J. Joseph, "Quantification of reflected wave magnitude and transit time using a multi-Rayleigh flow waveform model: A simplified approach to arterial wave separation analysis," *Biomed. Signal Process. Control*, vol. 93, no. July, p. 106129, 2024, doi: https://doi.org/10.1016/j.bspc.2024.106129.
- [27] E. Hermeling *et al.*, "Noninvasive assessment of arterial stiffness should discriminate between systolic and diastolic pressure ranges," *Hypertension*, vol. 55, no. 1, pp. 124–130, 2010, doi: 10.1161/HYPERTENSIONAHA.109.143867.
- [28] F. J. Callaghan, L. A. Geddes, C. F. Babbs, and J. D. Bourland, "Relationship between pulse-wave velocity and arterial elasticity," *Med. Biol. Eng. Comput.*, vol. 24, no. 3, pp. 248–254, 1986, doi: 10.1007/BF02441620.
- [29] R. Manoj, P. M. Nabeel, V. V. Abhidev, V. Raj Kiran, J. Joseph, and M. Sivaprakasam, "Demonstration of Pressure-Dependent Inter and Intra-Cycle Variations in Local Pulse Wave Velocity Using Excised Bovine Carotid Artery," *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS*, vol. 2020-July, pp. 2707–2710, 2020, doi: 10.1109/EMBC44109.2020.9175712.
- [30] K. V. Raj *et al.*, "High-frame-rate A-mode ultrasound for calibration-free cuffless carotid pressure: feasibility study using lower body negative pressure intervention," *Blood Press.*, vol. 31, no. 1, pp. 19–30, 2022, doi: 10.1080/08037051.2021.2022453.